

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)**

CHARUSAT Campus, Changa – 388 421.

**M Sc (BIOCHEMISTRY)**

**SEMESTER I EXAMINATION (BACKLOG)**

**BC701: GENERAL BIOCHEMISTRY AND MICROBIOLOGY**

**Date: 26.04.2011**

**Time: 10.00 AM to 1.00 PM**

**Total Marks: 70**

**Instructions to the candidates:**

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

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**PART A**

**Total marks: 20**

1. If  $\Delta G$  for a reaction is positive,
  - a) the forward reaction will tend to occur spontaneously.
  - b) the reverse reaction will tend to occur
  - c) both forward and reverse reactions occur at equal rates
  - d) none of the above
2. During electron transport in mitochondria proton motive force is generated due to  $H^+$  concentration gradient and ----- gradient.
3. The solubilization of solute in water is attributed to
  - (a) Hydrogen bond (b) covalent bond (c) Van der Waal's forces (d) dipoles
4. The weight of pure NaOH required to prepare 500ml of 0.1 N NaOH is
  - a) 4g b) 1g c) 10g d) 2g
5. Which of the following bonds are responsible for the backbone structure of proteins
  - a) Glycosidic bonds b) hydrogen bonds c) phosphodiester bonds d) peptide bonds
6. An organism is completely dependent on atmospheric  $O_2$  for growth. This organism is a(n)
  - a) osmotolerant. b) acidophile. c) facultative anaerobe d) obligate aerobe.
7. Increased concentration of salt and sugar in preservatives
  - a) Make them acid
  - b) Produce a hypotonic environment
  - c) Deplete nutrients
  - d) Produce a hypertonic environment
8. Cells are active and synthesizing new protoplasm. This stage of growth is called
  - a) Lag phase b) Stationary phase c) Log phase d) All of these
9. If the source of energy for an organism is from sunlight they are said to be
  - a) Phototrophs b) Heterotrophs c) Chemotrophs d) Chemolithotroph



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10. After a few generations microbial cells in culture enters a stationary phase due to factors such as  
a) nutrient limitation                      b) high density   c) increased temperature                      d) None of these
11. The functions of plasmid are  
a) DNA replication   b) Protein synthesis   c) Cell wall synthesis   d) None of the above
12. Anton Van Leeuwenhoek contributed ----- to the scientific world  
a) Stereoisomers   b) Fermentation   c) Microscope   d) Autoclave
13. Lysozyme is effective against  
a) Gram negative bacteria   b) Gram positive bacteria   c) Protozoa   d) Helminthes
14. In proteins, the phosphorylation is always at the amino acid residues  
a) aspartate or glutamate.  
b) glutamine or arginine.  
c) tyrosine, serine, or threonine.  
d) glycine, alanine, or aspartate.
15. Which of the following is plant lectin?  
a) Concanavalin A  
b) Mannose binding protein A  
c) Cholera toxin  
d) Enterotoxin
16. Which of the following statements about starch and glycogen is false?  
a) Both are homopolymers of glucose.  
b) Both serve primarily as structural elements in cell walls.  
c) Both are stored intracellularly as insoluble granules.  
d) Glycogen is more extensively branched than starch.

## State True or False:

17. Ion Transport may be coupled to a chemical reaction. (True/False)
18. Thus oligomycin & DCCD inhibit the ATP Synthase by interacting with Fo. (True/False)
19. If G is zero, both forward and reverse reactions occur at equal rates; the reaction is at equilibrium. (True/False)
20. Compounds with "high energy bonds" are said to have high group transfer potential. (True/False)



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**Part B**

**Total Marks: 50**

**Section I**

**Q.1 (A)** Give the difference between continuous and batch culture.

OR

**Q.1 (A)** Explain in brief synchronous culture.

(3)

**Q.1 (B)** Explain the following (any two)

- 1) phototrophs
- 2) heterotrophs
- 3) chemotrophs

OR

**Q.1 (B)** What are the important components of microbial culture media

(4)

**Q.1 (C)** Give different ways to control the growth of microorganisms.

(3)

**Q.2 (A)** Differentiate the bacterial cell on the basis of their cell wall, explaining the structure of cell wall. (4)

OR

**Q.2 (B)** Write the contribution of Anton Van Leeuwenhoek and Louise Pasteur in Microbial world. (4)

**Q.2 (C)** Describe in detail (Any THREE):

(6)

- i) Lipopolysaccharide
- ii) Capsule
- iii) Bacterial spores
- iv) Functions of cytoplasmic membrane



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## PART A

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